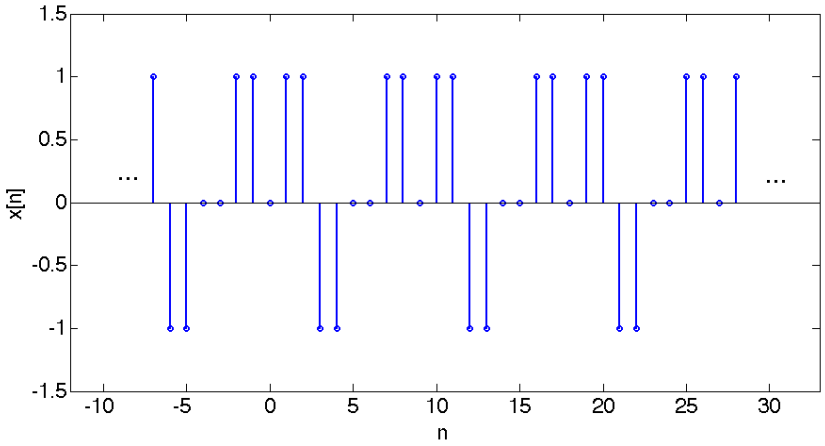
Signals: The Basics

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1. Consider the following signal.



This is a

a) discrete-time signal

b) continuous-time signal

and is

a) non periodic

b) periodic with fundamental period $N = 4$

c) periodic with fundamental period $N = 5$

d) periodic with fundamental period $N = 6$

e) periodic with fundamental period $N = 7$

f) periodic with fundamental period $N = 8$

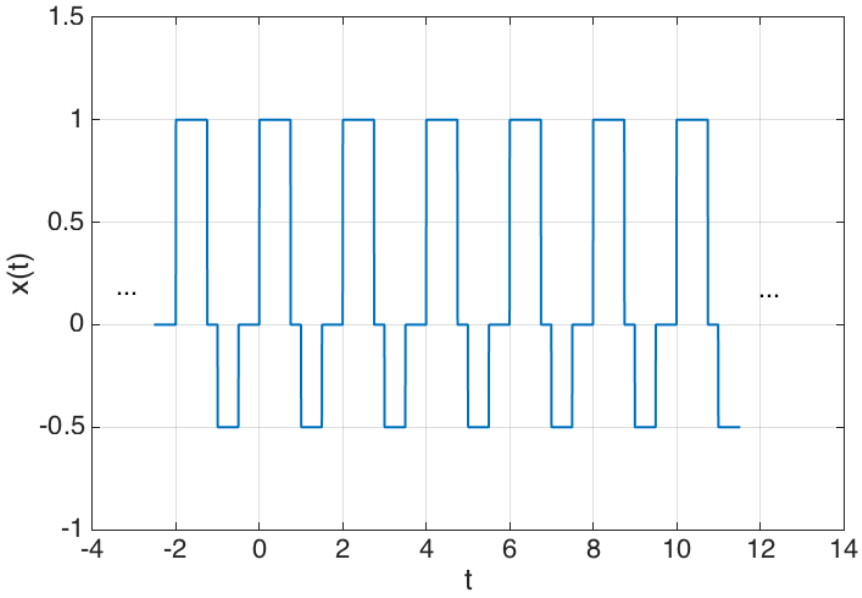
g) periodic with fundamental period $N = 9$

h) periodic with fundamental period $N = 10$

i) periodic with fundamental period $N = 11$

j) periodic with fundamental period $N = 12$

2. Consider the following signal.



This is a

a) discrete-time signal

b) continuous-time signal

and is

a) non periodic

b) periodic with fundamental period 1 second

c) periodic with fundamental period 2 second

d) periodic with fundamental period 3 second

e) periodic with fundamental period 4 second

f) periodic with fundamental period 5 second

g) periodic with fundamental period 6 second

h) periodic with fundamental period 7 second

i) periodic with fundamental period 8 second

3. The signal $x[n]$ is defined as

$$x[n] = \sin^2(\pi n/5)$$

This signal is

a) non periodic

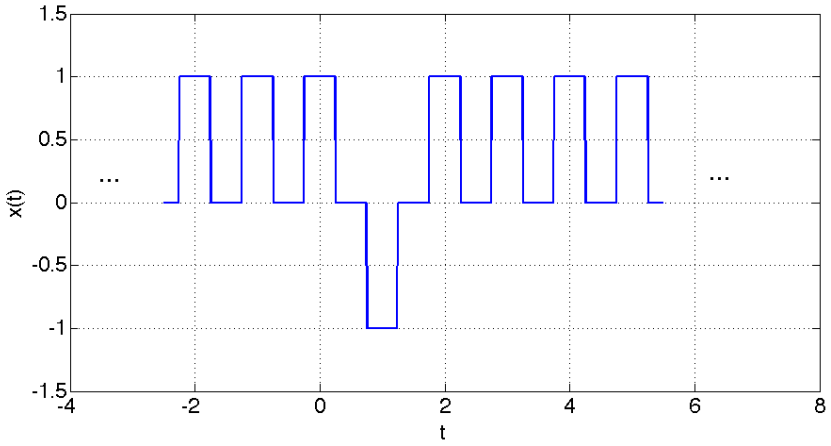
b) periodic

4. The signal $y[n]$ is defined as

$$y[n] = \cos^2(\pi n/6)$$

Find the fundamental period of $y[n]$.

5. Consider the signal $x(t)$ shown below. This signal is



a) non periodic

b) periodic with fundamental period 1 second

c) periodic with fundamental period 2 second

d) periodic with fundamental period 3 second

e) periodic with fundamental period 4 second

f) periodic with fundamental period 5 second

g) periodic with fundamental period 6 second

h) periodic with fundamental period 7 second

i) periodic with fundamental period 8 second

6. Use MATLAB code CTSinusoids.m to generate a graph of a sinusoid

$$x(t) = A\cos(2\pi ft + \phi)$$

on $0 \leq t \leq 10$ for the following values of A, f, and ϕ. In each case find the fundamental period.

a) $A = 2,\ f = 0.5,\ \phi = 0$

b) $A = 4,\ f = 1.25,\ \phi = 0$

c) $A = 4,\ f = 0.5,\ \phi = \pi/4$

d) $A = 2,\ f = 0.5,\ \phi = \pi/2$